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In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

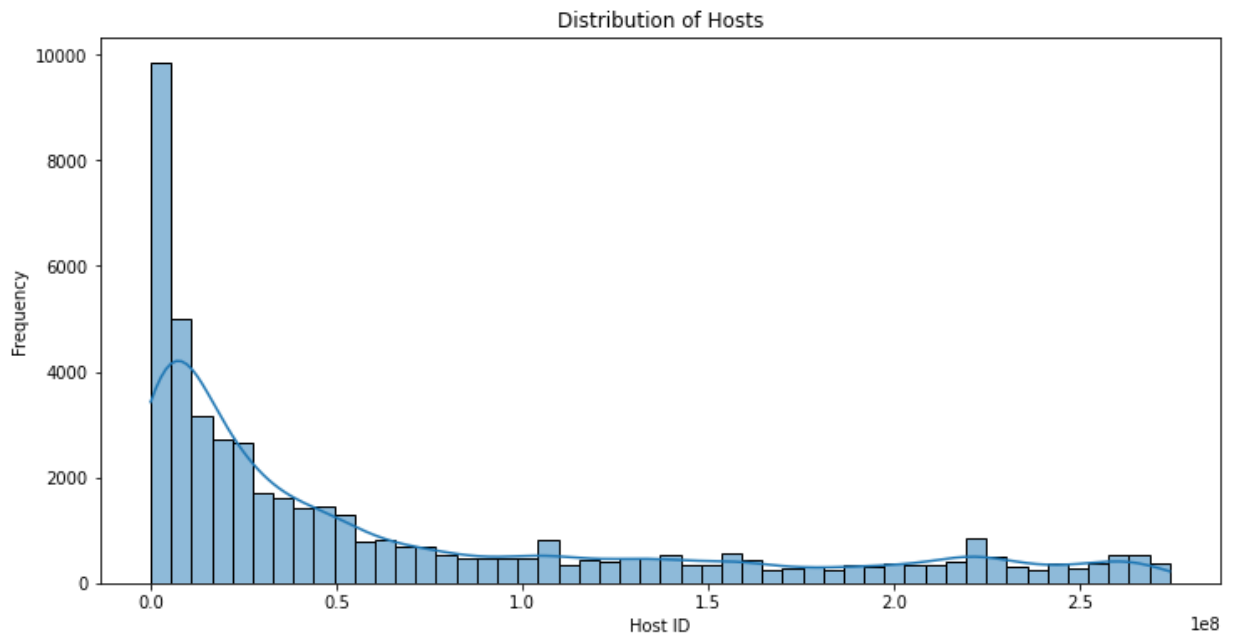
file_path = 'AB_NYC_2019.csv'
df = pd.read_csv(file_path)
```

```
In [ ]: df.head(5)
```

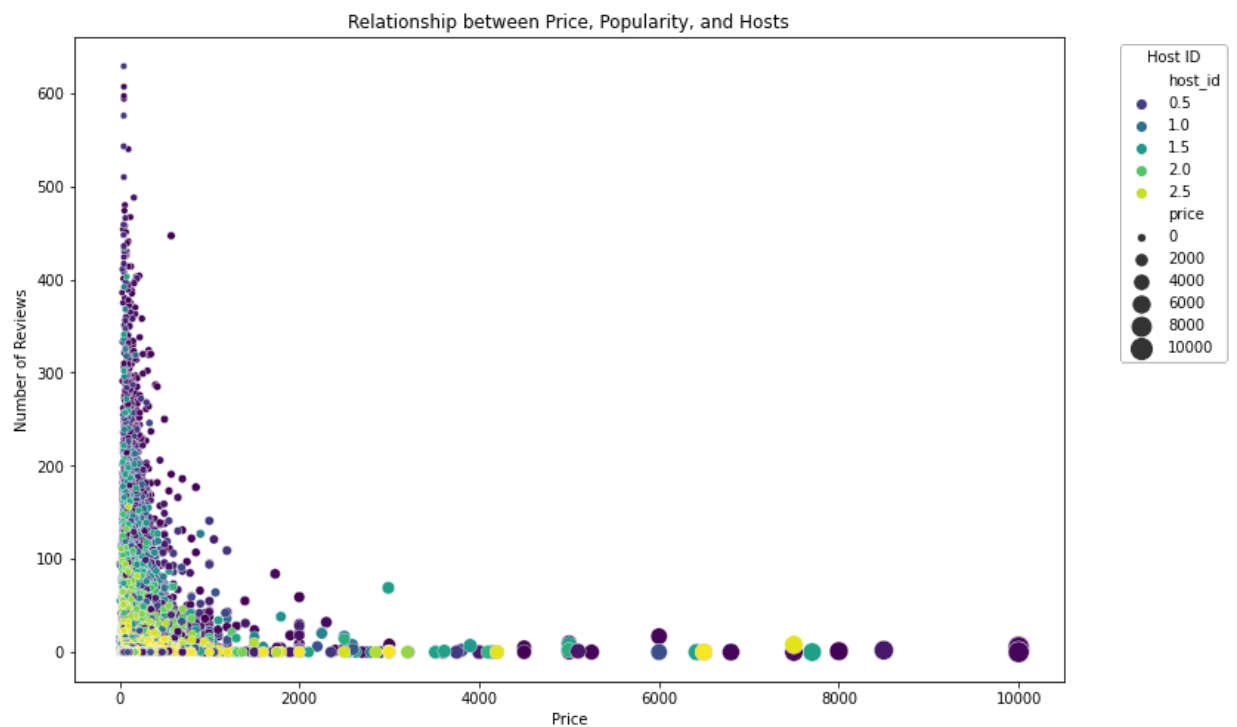
```
Out[ ]:
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	lat
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.81
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.81

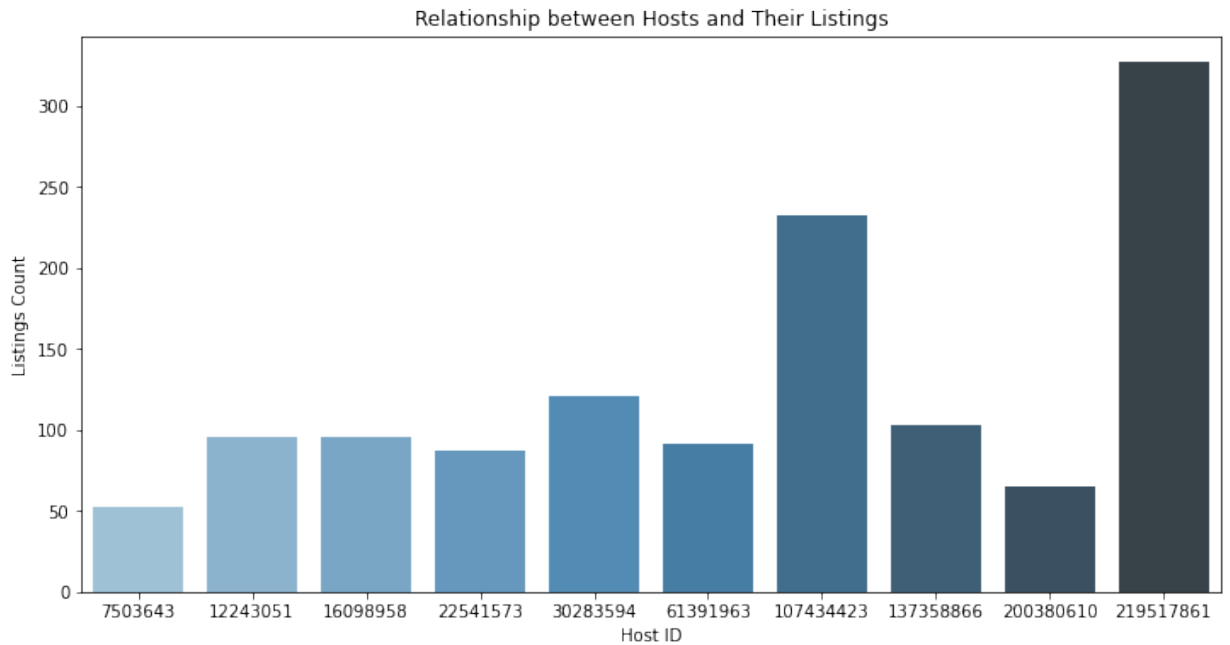
```
In [ ]: # Visualization 1: Distribution of Hosts
plt.figure(figsize=(12, 6))
sns.histplot(df['host_id'], bins=50, kde=True)
plt.title('Distribution of Hosts')
plt.xlabel('Host ID')
plt.ylabel('Frequency')
plt.show()
```



```
In [ ]: # Scatter plot for price vs. number of reviews with host differentiation
plt.figure(figsize=(12, 8))
sns.scatterplot(x='price', y='number_of_reviews', data=df, hue='host_id',
plt.title('Relationship between Price, Popularity, and Hosts')
plt.xlabel('Price')
plt.ylabel('Number of Reviews')
plt.legend(title='Host ID', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



```
In [ ]: # Bar plot for the top hosts with the highest count of listings
plt.figure(figsize=(12, 6))
listings_per_host = df['host_id'].value_counts().head(10).reset_index()
listings_per_host.columns = ['host_id', 'calculated_host_listings_count']
sns.barplot(x='host_id', y='calculated_host_listings_count', data=listings_per_host)
plt.title('Relationship between Hosts and Their Listings')
plt.xlabel('Host ID')
plt.ylabel('Listings Count')
plt.show()
```



```
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import seaborn as sns

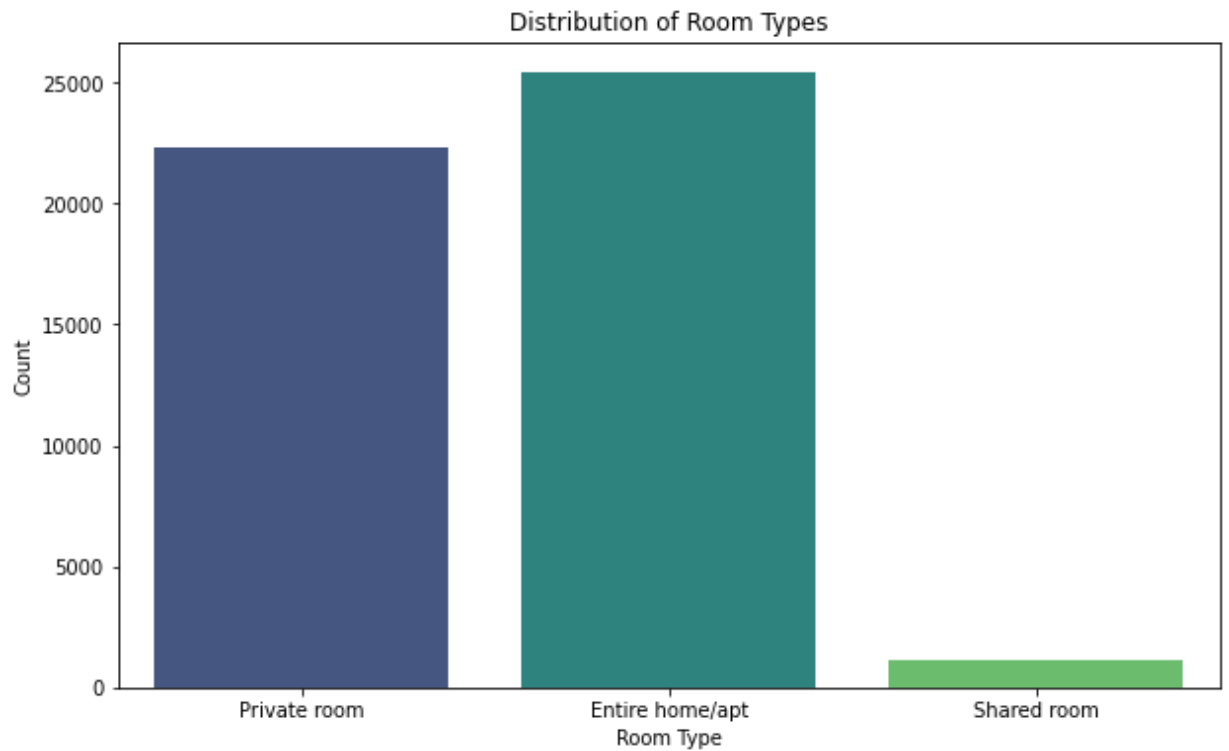
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Out[ ]:
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4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.81

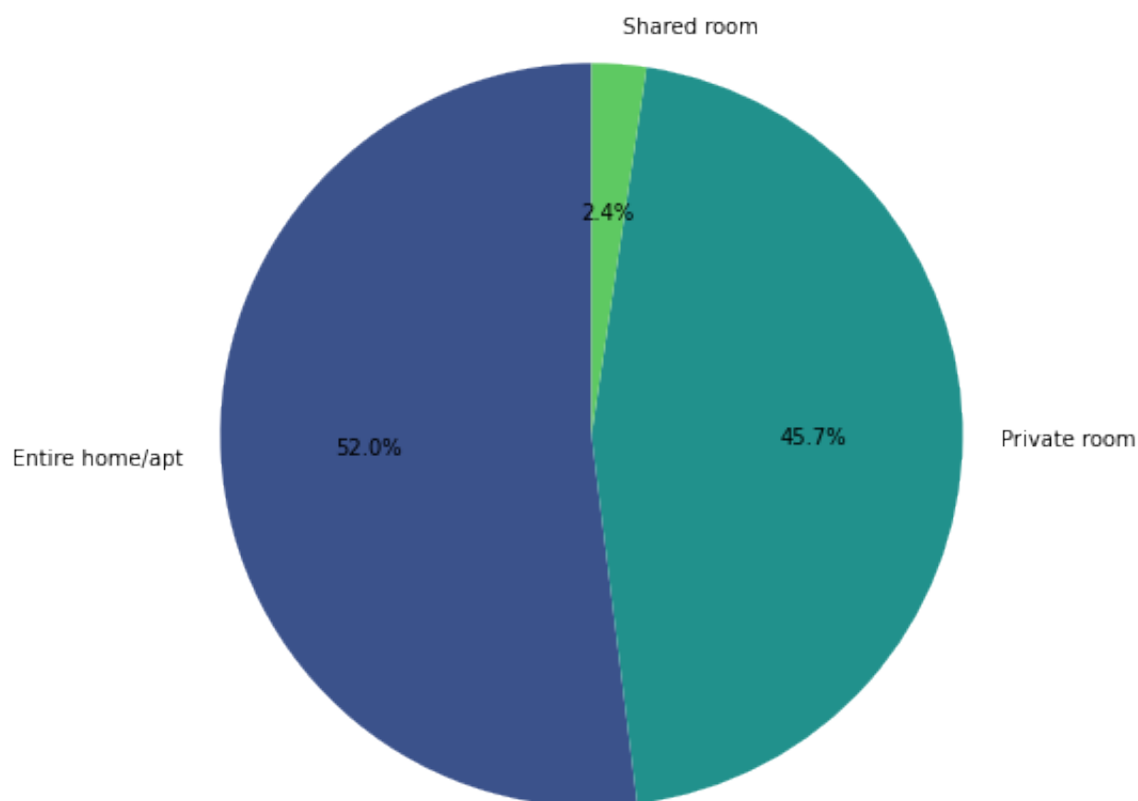
```
In [ ]: # Visualization: Distribution of Room Types
plt.figure(figsize=(10, 6))
sns.countplot(x='room_type', data=df, palette='viridis')
plt.title('Distribution of Room Types')
plt.xlabel('Room Type')
plt.ylabel('Count')
plt.show()
```



```
In [ ]: room_type_counts = df['room_type'].value_counts()
room_type_proportions = room_type_counts / room_type_counts.sum()
```

```
In [ ]: # Visualization 1: Pie Chart
plt.figure(figsize=(8, 8))
plt.pie(room_type_proportions, labels=room_type_proportions.index, autopct=
        colors=sns.color_palette('viridis', len(room_type_proportions)))
plt.title('Proportion of Room Types (Pie Chart)')
plt.show()
```

Proportion of Room Types (Pie Chart)



```
In [ ]: # Visualization: Violin Plot for Price Distribution by Room Type
plt.figure(figsize=(10, 6))
sns.violinplot(x='room_type', y='price', data=df, palette='Pastell1')
plt.title('Price Distribution by Room Type')
plt.xlabel('Room Type')
plt.ylabel('Price')
plt.show()
```

